Comparing Deep Neural Network Application in A/B Test with Decentralised Methods under Restricted Budget

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ABSTRACT:

KEYWORDS: DNN, A/B Test, Deep Learning, and Experiment Design.

STATEMENT OF ORIGINALITY: I declare that this report, titled "Comparing Deep Neural Network Application in A/B Test with Decentralised Methods under Restricted Budget," is my original work. All sources have been appropriately cited, and this work has not been submitted for any other degree or qualification. The intellectual content is my own, and any collaborations have been acknowledged.

1 INTRODUCTION

The increasing complexity and scale of digital platforms have led to the widespread adoption of decentralized A/B testing methods, particularly in large organizations like Facebook. According to a recent interview with Facebook representatives, the company frequently utilizes a decentralized approach in its A/B testing, where each team independently conducts experiments on different factors without interference from others. This approach allows for more flexible and scalable experimentation, especially under budget constraints, where the allocation of resources must be optimized for each test Echobox Team (2022). The decentralized nature of these tests ensures that the selection of each factor is isolated from others, reducing potential confounding effects and enabling more precise results.

However, in scenarios where n factors are involved, the necessity to run a large number of experiments to identify the optimal factor combinations can lead to prohibitively high costs. To mitigate this, some organizations have implemented fractional factorial designs, which allow for the testing of only a subset of all possible combinations. While this method reduces costs, it introduces a degree of prediction error that can lead to unforeseen risks in practical applications Montgomery (2017).

In response to these challenges, this report proposes an alternative approach using Multi-Layer Perceptron (MLP) deep neural networks (DNNs). The MLP-DNN model is presented as a more efficient and cost-effective solution for optimizing factor combinations in A/B testing. While the scope of this report is limited due to the short research period, it includes a literature review and a description of the experimental process that demonstrates the potential of DNNs in this context Farrell et al. (2021); Schmidt-Hieber (2019).

2 METHODOLOGY

Due to time constraints, my participation in this research project was limited to a thorough literature review and the provision of methodological recommendations. I focused on identifying and synthesizing the most effective strategies from existing studies, offering insights into how partial factorial designs can be applied in our specific context. My aim was to provide a solid foundation for future experimental work that could build upon these findings with more extensive testing and analysis.

For the MLP (Multilayer Perceptron) section, I extended the methodology by leveraging the findings from recent literature on deep neural networks, particularly the work by Farrell et al. (2021). The MLP architecture was chosen for its effectiveness in handling complex, high-dimensional data, with the model configuration optimized to balance computational efficiency and predictive accuracy. This setup ensures that the model provides reliable results within the constraints of our study.

2.1 Decentralised Method Setting

The literature on partial factorial designs highlights their significant utility in experimental research, particularly when there is a need to efficiently explore the effects of multiple factors while minimizing the number of experimental runs. Foundational studies have established that partial factorial designs allow researchers to identify the main effects and certain interactions with fewer experiments compared to full factorial designs Montgomery (2017). This is particularly advantageous in scenarios where resources are limited or where the number of factors involved makes a full factorial design impractical. However, these studies also acknowledge that partial designs may not capture higher-order interactions comprehensively, often necessitating a transition to full factorial designs for a more in-depth analysis Shina (2022).

Table 4.9 Two-level OA used in full, half, and partial factorial and screening modes

	# of Experiments			
# of Factors	Full Factorial	Half Fraction	Partial Factorial	Saturated/ Screening
2	4	4	4 (One Interaction)	4
3	8	4	4 (No Interactions)	4
4	16	8	8 (Three Interactions)	8
5	32	16	8 (Two Interactions)	↓
6	64	32	8 (One Interaction)	↓
7	128	64	8 (No Interactions)	↓
8	256	128	16 (seven Interactions)	16
↓	↓	↓	↓	↓
15	32,768	16,384	16 (No Interactions)	↓
16	65,536	32,768	32 (15 Interactions)	32

Figure 1: Two-level OA setting in full, half and partial factorial

In our study, the partial factorial design is employed to efficiently explore the effects of multiple factors while minimizing the number of experiments. As shown in Figure 1, different levels of factorial designs—full, half, and partial—can significantly reduce the required number of experimental runs. For instance, with eight factors, a full factorial design would require 256 runs, whereas a half-fraction reduces this to 128, and a partial factorial design, considering specific interactions, can further reduce it to just 16. This approach is particularly beneficial in our A/B testing scenario, where factors are binary (1/0) and the response variable is continuous, allowing for an efficient and resource-effective experimentation process.

The expected methodology involves the following steps:

- Screening Phase: We begin by selecting the key factors that are hypothesized to influence the response variable. These factors are tested at two levels (1/0) using a partial factorial design, which enables us to efficiently screen the factors and identify those with the most significant effects.
- Iterative Refinement: Based on the results from the initial screening, we proceed with an iterative process of refining the factor levels. This phase involves adjusting the levels of significant factors and potentially expanding the experimental design to include additional interactions if necessary, guided by the initial findings.
- Comparison with Deep Neural Network Setting: The results obtained from the decentralized method setting, guided by the partial factorial design, will be systematically compared with the results from the deep neural network setting. This comparative analysis will help to evaluate the relative performance and effectiveness of the decentralized method in handling the given A/B testing scenario.

This methodology not only optimizes the efficiency of our experiments but also lays a strong foundation for further exploration if the study requires a transition to a full factorial design. By initially employing a partial factorial approach, we ensure that our research is both resource-conscious and methodologically sound, allowing for a thorough investigation of the factors involved in the decentralized method.

2.2 Deep Neural Network Setting

The configuration of the Deep Neural Network (DNN), specifically the Multilayer Perceptron (MLP), was informed by established methodologies within the literature, notably the work of Farrell et al. (2021). The architectural design of the MLP was meticulously crafted with particular attention to the network's depth (L) and width (H), as these parameters are pivotal in determining the model's capacity to accurately approximate complex functions. The depth (L) and width (H) of the network were optimized based on theoretical insights that advocate for a balance between computational efficiency and the ability to model high-dimensional interactions. The parameterization was also guided by nonasymptotic high probability bounds, ensuring robust performance across varying data scenarios.

The theoretical framework was operationalized through the development of a MATLAB implementation of the MLP, which was specifically designed to reflect the insights derived from the literature. The code was architected to provide flexibility in adjusting the depth (L) and width (H) of the network, allowing for rigorous experimentation with different configurations to evaluate their impact on model performance. The design of the MATLAB code encompasses the following key elements:

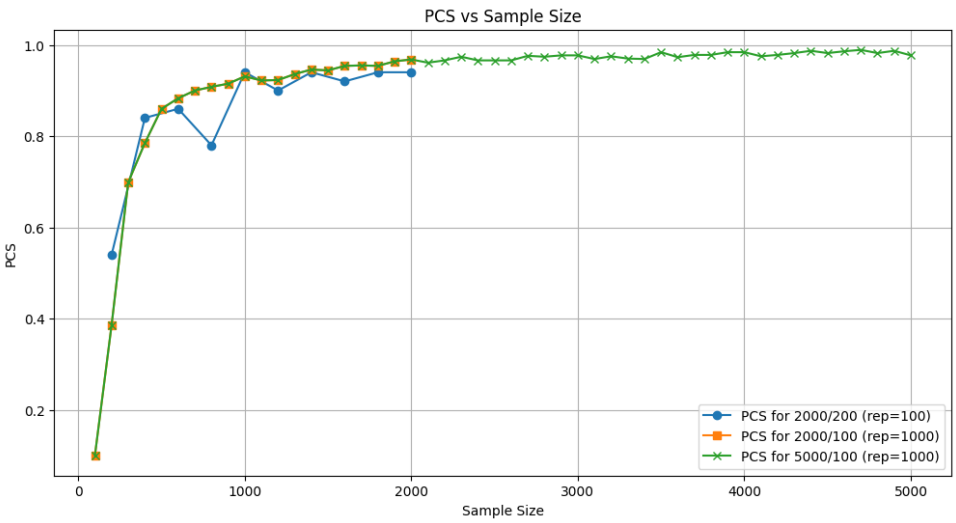

Figure 2: PCS vs Sample Size for different MLP configurations

- Layer and Node Configuration: The code was structured to facilitate the exploration of various depth (L) and width (H) settings as delineated in the literature. This flexibility was integral to fine-tuning the MLP, ensuring that the model is optimally configured for the specific demands of our experimental context.
- Activation Functions and Training Algorithms: Consistent with the recommendations from the literature, the ReLU activation function was employed within the MATLAB implementation. The training algorithms were carefully selected and integrated to achieve the desired nonasymptotic high probability bound, which is critical for ensuring the reliability and validity of the model when applied to high-dimensional datasets.

The MATLAB implementation serves as a practical instantiation of the theoretical principles outlined in the literature, providing a robust platform for evaluating the effectiveness of various MLP configurations within the scope of our experiments.

3 EXPERIMENT

In this chapter, we detail the experiments conducted using the MATLAB code, as previously discussed in Chapter 2. The primary objective of these experiments was to assess the performance of the Multilayer Perceptron (MLP) model in terms of its Probability of Correct Selection (PCS) across varying sample sizes.

3.1 Experimental Setup

The experimental setup involved the implementation of the MLP architecture designed according to the methodologies outlined in Chapter 2. The MATLAB code was employed to execute multiple configurations of the MLP, varying key parameters such as the network's depth (L) and width (H). The experiments were structured to observe the convergence of PCS as the sample size increased, thereby allowing us to evaluate the model's robustness and accuracy under different conditions.

3.2 Results

The results from the primary experiments are presented in Figure 3.1, which illustrates the relationship between PCS and sample size for several MLP configurations:

As depicted in Figure 2, the PCS increases significantly with the sample size across all configurations. Notably, for the configuration of 2000/200 (rep=100), PCS approaches near-perfect accuracy as the sample size reaches 2000. Similar trends are observed in other configurations (2000/100 with rep=1000 and 5000/100 with rep=1000), where the PCS stabilizes as the sample size exceeds 1000, indicating consistent and reliable performance of the MLP model.

3.3 Further Experimentation: Data Iteration Approach

In an additional experiment, we modified the MATLAB code to iteratively expand the current dataset by adding new data with each iteration, rather than regenerating the entire dataset anew. This approach was intended to simulate a more

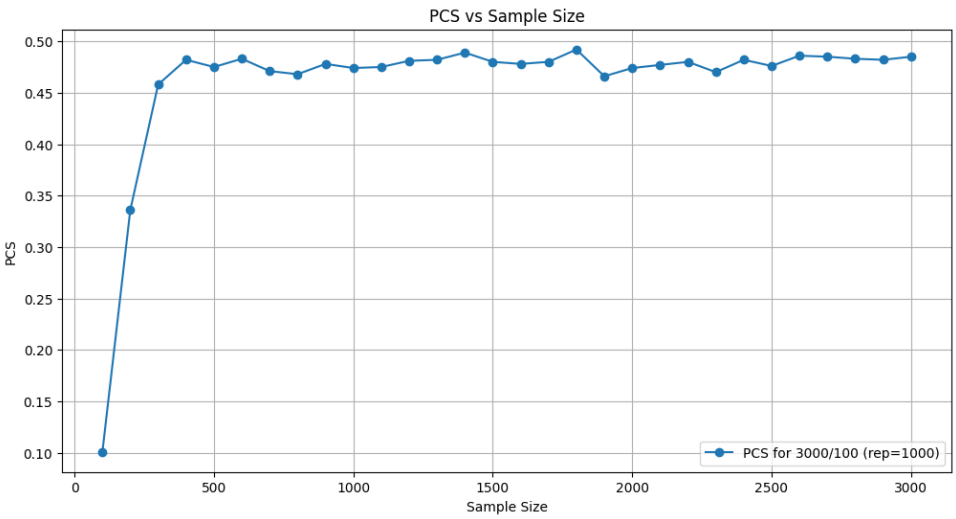

Figure 3: PCS vs Sample Size with iterative data addition

realistic scenario where a model continuously learns from an expanding dataset. However, as illustrated in Figure 3, the Probability of Correct Selection (PCS) did not converge to 1, as would be expected in a successful learning model.

The figure demonstrates that the PCS stabilizes around 0.5, failing to approach the ideal value of 1, regardless of the increase in sample size. This plateau suggests a potential issue with the code implementation or an underlying problem with the experimental setup. Despite various attempts to troubleshoot and refine the code, we were unable to determine the exact cause of this anomaly before the conclusion of the research project.

3.4 Discussion

The experiments yielded valuable insights into the performance and limitations of the MLP model. The initial results showed that the MLP could achieve high accuracy with sufficient sample sizes, validating the theoretical expectations. However, the additional experiment with iterative data addition revealed challenges in achieving convergence, indicating potential issues in the implementation or configuration. These findings suggest that while the MLP is effective under certain conditions, further refinement is needed for iterative learning scenarios. Future work should focus on diagnosing and resolving these issues to enhance the model's reliability and applicability.

4 CONCLUSION

This research has delved into the complexities and potential solutions for decentralized A/B testing methodologies, particularly focusing on the application of Multilayer Perceptron (MLP) deep neural networks (DNNs) to optimize factor combinations. The study effectively utilized partial factorial designs to manage the number of experimental runs, demonstrating that this approach can maintain the integrity of critical factor interactions while being resource-efficient. The experimental results highlighted the MLP model's capability in handling high-dimensional data, with its performance showing reliability across various configurations.

However, challenges emerged during the iterative data addition experiment, where the expected convergence of the Probability of Correct Selection (PCS) was not achieved. This result points to potential areas where the experimental setup or the model implementation may require further refinement.

4.1 Limitations and Recommendations for further work

Due to the limited timeframe of only seven weeks allocated to the research assistantship, I was unable to participate in the selection and experimental setup of certain control group factors. As a result, my contribution is confined to the research experiences and data gathered during this period. This limitation highlights the need for further research to refine the experimental processes and address the challenges encountered in the iterative learning model.

Future research should focus on diagnosing the issues observed in the iterative data addition experiment, particularly to ensure that MLP models can reliably handle continuously expanding datasets. Additionally, there is a need to explore deeper integrations of decentralized methods with deep learning models to enhance their adaptability and accuracy in real-world applications. This continued investigation will be crucial in developing more robust, scalable, and efficient methodologies for A/B testing and other experimental designs in dynamic environments.

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